



Product Requirements Document

Orviohub Inventory MVP

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Product owner: Code X

Product category: Inventory, sales, and basic business management SaaS

Initial market: Nigerian micro and small businesses

Initial launch location: Enugu, Nigeria

Product status: MVP definition

1. Executive summary

Orviohub Inventory is a simple, affordable, Nigerian-focused inventory and sales management product for businesses that currently manage stock and sales using notebooks, spreadsheets, memory, WhatsApp conversations, or basic cash registers.

The product will help merchants:

- Create and manage products.
- Track stock.
- Record purchases.
- Record sales.
- Receive payments.
- Track customer credit.
- Manage suppliers.
- Monitor low-stock items.
- Record stock losses and adjustments.
- View daily and historical reports.
- Operate basic sales workflows during poor internet connectivity.

The long-term vision is to develop an Orviohub business platform with multiple modules similar to Odoo. However, this MVP will focus strictly on the business loop:

Product → Stock received → Sale → Payment → Receipt → Stock deduction → Report → Reorder

The MVP should prove that Nigerian businesses will consistently use and pay for a digital inventory system before more advanced modules are built.

2. Product vision

Vision statement

Help Nigerian businesses replace unreliable manual records with simple, accessible, and trustworthy digital tools.

Product promise

Know what you have, what you sold, what you bought, what customers owe, and what to reorder.

Long-term ambition

Orviohub may eventually become a modular business operating system containing:

- Inventory.
- Sales.
- Accounting.
- CRM.
- Expenses.
- Invoicing.
- Staff management.
- Gym management.
- Bookings.
- Subscriptions.
- E-commerce.
- Field service.

The MVP must establish reusable platform foundations without allowing the long-term vision to expand the initial product beyond what can be properly built and supported.

3. Background and opportunity

Many Nigerian small businesses do not require a complex ERP at the beginning. They need simple answers to immediate operational questions:

- How many units are left?
- Which product sold today?
- Which worker made the sale?

- How much cash was collected?
- How much came through bank transfer?
- Which customers owe money?
- Which suppliers have not been paid?
- Which products need restocking?
- Did stock disappear or was it recorded incorrectly?

The Nigerian market already includes inventory and POS competitors with features such as offline operation, stock management, supplier ledgers, Naira pricing, and local support.

Examples include MyTreda, InkeepX, Tracepos, App4Stores, Vendloop, and related products. ^[1]
^[2] ^[3] ^[4]

This confirms that the market problem is real, but it also means Orviohub needs a clear position. It should not compete only by claiming to have “many features.”

Recommended market position

A simple, offline-capable inventory and sales system for Nigerian retailers.

The product should compete on:

- Ease of use.
- Reliable stock records.
- Offline sales.
- Naira-first workflows.
- WhatsApp-friendly receipts.
- Affordable pricing.
- Assisted onboarding.
- Local support.
- Clear reporting.

4. Problem statement

Current customer problems

Small-business owners frequently experience:

- Inaccurate stock counts.
- Unrecorded sales.
- Staff-related leakage.
- Forgotten debts.
- Unclear profit.

- Slow end-of-day calculations.
- Poor supplier tracking.
- Manual stock-counting errors.
- No reliable sales history.
- Poor visibility across branches.
- Difficulty operating during network failure.
- Difficulty transitioning existing records into software.

Problem impact

These issues can cause:

- Lost revenue.
- Overstocking.
- Stockouts.
- Cash discrepancies.
- Poor purchasing decisions.
- Customer dissatisfaction.
- Time wasted preparing reports.
- Conflict between owners and staff.
- Inability to determine whether the business is profitable.

Product opportunity

Orviohub can create value by making the most important daily activities faster and more reliable:

- Recording a sale.
- Deducting stock.
- Recording a purchase.
- Tracking credit.
- Checking low stock.
- Reviewing daily sales.

5. Product goals

Business goals

- Validate demand for Nigerian inventory software.
- Acquire an initial pilot group in Enugu.
- Convert active pilot users into paying customers.
- Build a repeatable onboarding process.
- Create a reusable multi-tenant platform foundation.
- Build trust around accurate transaction and stock records.
- Learn which additional modules customers need most.

Customer goals

Customers should be able to:

- Know current stock.
- Record every sale.
- View daily sales.
- Track payment methods.
- Monitor customer debts.
- Record supplier purchases.
- Identify low-stock products.
- Review basic profit estimates.
- Understand staff activity.
- Continue selling during temporary internet failure.
- Export their business records.

Product goals

- First sale completed within the onboarding session.
- Common POS sale completed quickly.
- Stock-changing actions fully traceable.
- Offline transactions synchronized safely.
- Reports understandable without accounting expertise.
- Application usable on low-cost Android devices.
- Product setup possible without technical assistance.

6. Non-goals

The MVP will not attempt to become a complete ERP.

The following are not included in the initial scope:

- Full double-entry accounting.
- Payroll.
- Manufacturing.
- Bills of materials.
- Advanced warehouse automation.
- Advanced CRM.
- Marketing automation.
- Full e-commerce marketplace.
- Gym management.
- Booking management.
- Complex subscription management.
- AI demand forecasting.
- Advanced route planning.
- Full enterprise workflow approvals.
- Native iOS application.
- Complete tax advisory.
- Full e-invoicing implementation without validated regulatory and technical requirements.
- Highly customized per-customer workflows.
- Microservices architecture.

7. Target customers

Primary customer profile

- Nigerian small retailer or wholesaler.
- One to three shops or branches.
- Physical inventory.
- Approximately 100–5,000 products.
- Uses notebooks, Excel, or informal records.
- Has one to ten staff members.
- Requires basic sales and stock visibility.
- May experience unstable connectivity.

- Wants affordable local software.

Recommended launch segments

Provision stores and mini supermarkets

Why they are suitable:

- Frequent transactions.
- Large number of products.
- Repeated stock purchases.
- Clear need for low-stock alerts.
- Common use of cash, transfer, POS, and credit.

Boutiques and fashion retailers

Important product requirements:

- Size variants.
- Colour variants.
- Product images.
- Multiple pricing levels.
- Returns.

Cosmetics shops

Important product requirements:

- Product variants.
- Expiry awareness in later versions.
- Supplier tracking.
- Small-item barcode scanning.

Electronics and phone accessories

Important product requirements:

- Product variants.
- Serial numbers in a later release.
- Warranty tracking in a later release.
- Supplier records.
- Product images.

Segments for later

Pharmacies

Pharmacy workflows may require:

- Batch numbers.
- Expiry dates.
- Controlled-product records.
- Recall support.
- More advanced compliance.

Restaurants

Restaurant inventory requires:

- Recipes.
- Ingredients.
- Modifiers.
- Wastage.
- Kitchen operations.

Manufacturers

Manufacturing requires:

- Bills of materials.
- Work orders.
- Raw materials.
- Production costing.
- Quality control.

These should not shape the first release.

8. User personas

Persona A: Business owner

Example: A provision-store owner in Enugu.

Goals

- Understand stock position.
- Know daily sales.
- Reduce theft and loss.
- Monitor staff.
- Track debtors.
- Know which products are profitable.
- Make better purchasing decisions.

Frustrations

- Manual notebooks.
- Inconsistent staff records.
- Stock disappearing.
- Too much time spent calculating sales.
- Unclear business profit.
- Fear of losing digital records.
- Limited time to learn software.

Important product characteristics

- Simple screens.
- Fast onboarding.
- Daily summary.
- Data backup.
- Mobile access.
- Easy support.

Persona B: Sales attendant

Goals

- Complete sales quickly.
- Search or scan products.
- Receive payment.
- Issue receipts.

Restrictions

- Cannot see cost prices by default.
- Cannot change product prices without permission.
- Cannot adjust stock.
- Cannot delete sales.
- Cannot access owner-only reports.

Persona C: Stock manager

Goals

- Receive purchases.
- Count stock.
- Track discrepancies.
- Record damaged products.
- Monitor reorder levels.

Persona D: Accountant or bookkeeper

Goals

- Review transaction history.
- Export reports.
- Track customer and supplier balances.
- Compare sales and purchases.
- Estimate profit.

Persona E: Platform administrator

Goals

- Manage organizations.
- Monitor usage.
- Manage subscriptions.
- Support customers.
- Investigate errors.
- Monitor synchronization failures.

9. Jobs to be done

Job 1: Record a sale

When I sell a product, I want to record it quickly so stock and revenue stay accurate.

Job 2: Check stock

When I need to know whether an item is available, I want to search quickly so I can answer the customer immediately.

Job 3: Receive stock

When goods arrive from a supplier, I want to record the received quantity and cost so stock and supplier balances are updated.

Job 4: Track customer debt

When I sell on credit, I want to record the amount owed so I do not rely on memory.

Job 5: Identify low stock

When inventory is getting low, I want to be notified so I can reorder before a stockout.

Job 6: Operate offline

When the network is unavailable, I want to continue selling and synchronize later without losing records.

Job 7: Review business performance

At the end of the day, I want to know how much I sold, how I was paid, and whether stock records are complete.

10. Product principles

Simple by default

The system should expose only what users need for the current task.

Mobile-first

The most important features should work well on Android phones and tablets.

Offline where it matters

Sales, product lookup, receipt creation, and local stock deduction should be available offline.

No silent stock changes

Every change must have a source and reason.

Human support is part of the product

Assisted onboarding and responsive support are especially important for first-time software users.

Upgrade-safe architecture

Custom behavior should be built through modules and extension points rather than frequent direct modification of core logic.

Transparent pricing

Customers should understand what is included, what is limited, and what costs extra.

11. MVP scope

Included

- Account registration.
- Organization setup.
- Branch setup.
- User invitations.
- Roles and permissions.
- Product catalogue.
- Categories.
- Barcode support.
- Opening stock.
- Stock movement ledger.
- Sales/POS.
- Cash, transfer, POS, USSD, and credit payment recording.
- Purchases.
- Suppliers.
- Customers.
- Credit sales.

- Customer payments.
- Returns.
- Stock counts.
- Stock adjustments.
- Low-stock alerts.
- Basic expenses.
- Reports.
- CSV import/export.
- Offline sales.
- Synchronization.
- Audit logs.
- Subscription plans.
- Basic support tools.

12. Detailed functional requirements

12.1 Authentication

Requirements

- User can register with email or phone.
- User can log in.
- User can log out.
- User can reset password.
- User can verify contact information.
- User can change password.
- User can manage active sessions.
- Owner can deactivate users.
- Login attempts are rate-limited.
- Authentication events are logged.

Acceptance criteria

- Unauthenticated users cannot access business data.
- Deactivated users cannot log in.
- Password-reset links expire.
- Passwords are not stored in plain text.

- Session revocation works.

12.2 Organization setup

Requirements

The owner can:

- Create an organization.
- Enter business name.
- Enter business phone number.
- Enter address.
- Add logo.
- Select business category.
- Configure currency.
- Create primary branch.
- Configure receipt footer.
- Configure tax display.
- Invite staff.
- Select a subscription plan.

Onboarding flow

1. Register.
2. Create business.
3. Select business type.
4. Add first branch.
5. Add or import products.
6. Enter opening stock.
7. Invite staff.
8. Record first sale.
9. View dashboard.

Acceptance criteria

- Business can be created in one session.
- Primary branch is created automatically.
- NGN is the default currency.
- Owner is automatically assigned the Owner role.

- First sale can be completed after product setup.

12.3 Roles and permissions

Initial roles

- Owner.
- Manager.
- Sales attendant.
- Stock manager.
- Accountant.

Permissions

- View products.
- Create products.
- Edit products.
- Archive products.
- View stock.
- Adjust stock.
- Receive purchases.
- Record sales.
- Cancel sales.
- Process returns.
- View cost price.
- View profit.
- View reports.
- Export data.
- Manage users.
- Manage settings.
- Manage billing.

Default role behavior

Permission	Owner	Manager	Sales attendant	Stock manager	Accountant
Record sales	Yes	Yes	Yes	Optional	Optional
View cost prices	Yes	Configurable	No	Configurable	Yes
Adjust stock	Yes	Yes	No	Yes	No

Permission	Owner	Manager	Sales attendant	Stock manager	Accountant
Receive purchases	Yes	Yes	No	Yes	Optional
View profit	Yes	Configurable	No	No	Yes
Export data	Yes	Configurable	No	Configurable	Yes
Manage users	Yes	Configurable	No	No	No
Manage billing	Yes	No	No	No	No

Security requirement

Every permission must be checked on the server.

12.4 Product catalogue

Product fields

- Product ID.
- Product name.
- SKU.
- Barcode.
- Category.
- Brand.
- Description.
- Image.
- Base unit.
- Purchase cost.
- Retail price.
- Wholesale price.
- Reorder level.
- Preferred supplier.
- Tax status.
- Active status.
- Created by.
- Created date.
- Updated date.

Requirements

- Create product.
- Edit product.
- Archive product.
- Restore product.
- Search product.
- Filter product.
- Sort product.
- Scan barcode.
- Add image.
- Import CSV.
- Export CSV.
- Assign multiple prices.
- Warn about duplicate SKU.
- Prevent duplicate barcode.
- Preserve historical product references.

Product status

- Active.
- Archived.

Archived products must not appear in normal sales search but must remain available in historical transactions.

12.5 Categories and units

Initial categories

Users can create their own categories.

Examples:

- Beverages.
- Groceries.
- Cosmetics.
- Clothing.
- Electronics.
- Accessories.

- Household items.
- Building materials.

Initial units

- Piece.
- Pack.
- Box.
- Carton.
- Bottle.
- Bag.
- Kilogram.
- Gram.
- Litre.
- Millilitre.
- Metre.

Unit conversion

The data model should support conversions such as:

- One carton = 24 pieces.
- One box = 12 units.
- One pack = 6 units.

A basic conversion feature may be included in a later MVP iteration if it threatens the initial delivery timeline.

12.6 Barcode support

Requirements

- Scan barcode through phone camera.
- Support external scanner input.
- Search by barcode.
- Add barcode manually.
- Detect duplicate barcodes.
- Show product-not-found state.
- Create unknown product quickly.
- Scan products during sales.

- Scan during receiving.
- Scan during stock counts.

Barcode workflows are valuable for stock receiving, sales, and inventory operations; Odoo's documentation similarly supports barcode-based product lookup and inventory processing. ^[5]
^[6]

12.7 Inventory ledger

Stock movement types

- Opening stock.
- Purchase received.
- Sale.
- Customer return.
- Supplier return.
- Transfer out.
- Transfer in.
- Damage.
- Expiry.
- Loss.
- Theft.
- Manual increase.
- Manual decrease.
- Stock-count correction.

Movement fields

- Movement ID.
- Organization ID.
- Branch ID.
- Product ID.
- Quantity.
- Unit.
- Movement type.
- Reason.
- Related transaction.
- User ID.

- Device ID.
- Timestamp.
- Sync status.
- Reversal reference.

Requirements

- Show current quantity.
- Show quantity by branch.
- Show stock history.
- Show movement source.
- Show adjustment reason.
- Show user.
- Show timestamp.
- Prevent unauthorized adjustments.
- Configure negative-stock behavior.
- Display low-stock products.
- Preserve movement history.

Core rule

Stock quantity must be changed through a stock movement, not by direct silent editing.

12.8 Opening stock

Requirements

- User can enter opening stock manually.
- User can import opening stock by CSV.
- User can assign opening stock to a branch.
- User can provide opening cost.
- System creates opening-stock movements.
- Opening stock import creates an import log.
- Opening stock cannot be silently changed after posting.

12.9 Stock receiving

Flow

1. Select supplier.
2. Select purchase order or unscheduled receipt.
3. Scan or search product.
4. Enter quantity.
5. Enter actual cost.
6. Add invoice reference.
7. Review receipt.
8. Confirm.
9. Increase stock.
10. Update supplier balance.
11. Update cost history.

Requirements

- Full receiving.
- Partial receiving.
- Damaged quantity.
- Missing quantity.
- Draft receiving.
- Cancellation before confirmation.
- Supplier invoice reference.
- Payment amount.
- Amount owed.
- Receipt history.

Stock should increase only when the receipt is confirmed.

12.10 Stock adjustments

Requirements

- Authorized user starts adjustment.
- User selects product and branch.
- User enters adjustment quantity or final quantity.
- User selects reason.

- User adds note.
- User may attach image.
- System shows old quantity.
- System shows new quantity.
- System shows variance.
- User confirms.
- System creates movement.
- System writes audit log.

Mandatory adjustment reasons

- Damaged.
- Expired.
- Missing.
- Theft.
- Counting error.
- Opening correction.
- Other.

12.11 Stock counts

Requirements

- Start count session.
- Select branch.
- Filter by category.
- Scan products.
- Enter counted quantity.
- Save incomplete count.
- Review variance.
- Submit for approval.
- Confirm adjustment.
- Export count report.

Approval

Large discrepancies should require owner or manager approval.

12.12 Sales and POS

Sales flow

1. Open Sell.
2. Search or scan product.
3. Add item.
4. Set quantity.
5. Select customer.
6. Apply discount if permitted.
7. Select payment method.
8. Enter amount and reference.
9. Confirm payment.
10. Generate receipt.
11. Create stock movement.
12. Update customer balance.
13. Synchronize.

Required payment methods

- Cash.
- Bank transfer.
- POS terminal.
- USSD.
- Credit.
- Other.
- Split payment.

Requirements

- Add item.
- Remove item.
- Edit quantity.
- Search.
- Scan.

- Apply discount.
- Select customer.
- Add walk-in customer.
- Calculate change.
- Record transaction reference.
- Record partial payment.
- Record credit.
- Print receipt.
- Share receipt.
- Reprint receipt.
- Cancel sale.
- Return sale.
- Track salesperson.
- Work offline.
- Display sync state.

12.13 Offline sales

Offline capabilities

- Search cached products.
- Scan cached products.
- Complete sale.
- Record payment method.
- Record credit.
- Generate receipt.
- Deduct local stock.
- View local sales.
- Queue transaction for synchronization.

Offline transaction fields

- Local transaction ID.
- Device ID.
- User ID.
- Organization ID.
- Branch ID.

- Product lines.
- Prices.
- Payment information.
- Created timestamp.
- Sync status.
- Idempotency key.

Sync states

- Local only.
- Queued.
- Uploading.
- Synchronized.
- Failed.
- Conflict.
- Resolved.

Synchronization requirements

- Automatic retry.
- Exponential backoff.
- Duplicate prevention.
- Failed-record visibility.
- Manual retry.
- Conflict visibility.
- Sync timestamps.
- Local data retention until successful sync.

Conflict behavior

Conflicts may occur when:

- Two devices sell the last available unit offline.
- Product price changes online while a device is offline.
- A product is archived while a device has stale data.
- A sale is cancelled on one device while another device has not synchronized.

The system must not silently hide or delete these situations.

12.14 Returns

Customer returns

- Find original sale.
- Select returned lines.
- Select return reason.
- Specify resalable or damaged status.
- Add items back to stock if resalable.
- Record refund or customer credit.
- Generate return receipt.
- Require permission.
- Preserve original sale.

Supplier returns

- Reference original purchase.
- Select items.
- Reduce stock.
- Update supplier balance.
- Record reason.
- Preserve purchase history.

12.15 Purchases

Purchase states

- Draft.
- Ordered.
- Partially received.
- Fully received.
- Cancelled.
- Closed.

Requirements

- Create purchase order.
- Add supplier.
- Add products.

- Enter expected quantities.
- Enter expected costs.
- Record payment.
- Record supplier debt.
- Receive all or part of order.
- View purchase history.
- Export purchase data.

12.16 Suppliers

Supplier fields

- Name.
- Phone number.
- Email.
- Address.
- Contact person.
- Tax information where required.
- Opening balance.
- Notes.
- Active status.

Requirements

- Create supplier.
- Edit supplier.
- Archive supplier.
- View supplier products.
- View purchase history.
- View balance.
- Record payment.
- View supplier statement.
- Export supplier data.

12.17 Customers and credit

Customer fields

- Name.
- Phone number.
- Email.
- Address.
- Credit limit.
- Opening balance.
- Notes.
- Active status.

Requirements

- Create customer.
- Search customer.
- Record credit sale.
- Record partial payment.
- Record full payment.
- View balance.
- View payment history.
- View purchases.
- Generate statement.
- Share statement.
- Record payment note.
- Set credit limit.
- Block credit when limit is exceeded if enabled.

Balance terminology

Use clear terms:

- Sale total.
- Paid.
- Due.
- Overdue.

12.18 Expenses

Scope

Expenses are included only as lightweight business records.

Requirements

- Create expense.
- Add category.
- Add amount.
- Add payment method.
- Add date.
- Add note.
- Attach receipt.
- Edit before finalization.
- Restrict deletion.
- Include in cash summary.
- Export.

Expense categories

- Rent.
- Transport.
- Electricity.
- Internet.
- Repairs.
- Packaging.
- Delivery.
- Staff welfare.
- Miscellaneous.

The product should not describe this as full accounting.

12.19 Receipts

Receipt fields

- Business name.
- Logo.
- Branch.
- Address.
- Phone number.
- Receipt number.
- Date.
- Time.
- Sales attendant.
- Customer.
- Items.
- Quantity.
- Unit price.
- Discount.
- Tax.
- Total.
- Amount paid.
- Amount due.
- Payment method.
- Transaction reference.
- Footer message.

Receipt delivery

- Print.
- Download.
- Share.
- Copy text.
- Send through supported WhatsApp workflow.
- Reprint.
- Resend.

12.20 Reports

Dashboard

The owner dashboard should show:

- Today's sales.
- Number of transactions.
- Cash received.
- Transfer received.
- POS received.
- Credit sales.
- Estimated gross profit.
- Low-stock count.
- Customer debt.
- Supplier debt.
- Pending sync count.
- Recent adjustments.

Required reports

- Sales by date.
- Sales by product.
- Sales by category.
- Sales by staff.
- Sales by payment method.
- Purchases.
- Stock on hand.
- Stock valuation.
- Low stock.
- Stock movements.
- Customer debt.
- Supplier balance.
- Returns.
- Expenses.
- Profit estimate.
- Stock adjustments.

Filters

- Date.
- Branch.
- Product.
- Category.
- Customer.
- Supplier.
- Staff.
- Payment method.
- Status.

Report requirements

- Use consistent totals.
- Exclude cancelled transactions.
- Reflect returns correctly.
- Display report generation time.
- Show whether data includes unsynchronized transactions.
- Export CSV.
- Provide print-friendly view.

12.21 Profit calculation

The MVP may show estimated gross profit.

\$\$ Estimated gross profit

Net sales

Estimated cost of goods sold

\$\$

The product must document:

- Costing method.
- Treatment of discounts.
- Treatment of returns.
- Treatment of taxes.

- Treatment of damaged stock.
- Treatment of expenses.
- Whether unsynchronized sales are included.

Odoo's inventory documentation describes valuation approaches such as standard cost, average cost, and FIFO. ^[7]

Recommended MVP method

Use weighted average cost if the implementation timeline allows it. Otherwise, use last purchase cost and clearly label results as estimated.

12.22 Low-stock alerts

Requirements

- Product has reorder level.
- System compares stock with reorder level.
- Dashboard shows low-stock products.
- Owner can receive alert.
- User can filter low-stock items.
- User can mark alert as reviewed.
- User can initiate a purchase draft in a future release.

Future enhancement

Recommended reorder quantities can later use:

- Average sales velocity.
- Supplier lead time.
- Seasonal demand.
- Minimum order quantity.
- Safety stock.

12.23 Multi-branch support

MVP requirements

- Create branch.
- Edit branch.
- Archive branch.

- Assign users.
- Assign products and stock.
- Record sales by branch.
- Record purchases by branch.
- Filter reports by branch.

Later MVP requirements

- Transfer stock.
- Branch-specific permissions.
- Branch performance comparison.
- Consolidated reports.

12.24 Import and export

Import

Support:

- Products.
- Opening stock.
- Customers.
- Suppliers.
- Customer balances.
- Supplier balances.

Import flow

1. Download template.
2. Upload CSV.
3. Validate file.
4. Display row-level errors.
5. Preview.
6. Confirm.
7. Process import.
8. Show summary.
9. Save import log.

Export

Support:

- Products.
- Sales.
- Purchases.
- Stock movements.
- Customers.
- Suppliers.
- Expenses.
- Payments.
- Reports.

Exports must be permission-controlled, organization-scoped, and logged.

12.25 Notifications

Notifications

- Low stock.
- Failed synchronization.
- Unresolved conflict.
- Customer payment reminder.
- Supplier payment reminder.
- Daily sales summary.
- Staff invitation.
- Subscription expiry.
- Failed subscription payment.
- Stock adjustment summary.

Channels

- In-app.
- Email.
- WhatsApp where officially supported.
- SMS for selected critical notices.

12.26 Nigerian localization

Requirements

- NGN as default currency.
- Nigerian phone-number formats.
- Local date and number formatting.
- Cash, transfer, POS, and USSD payment labels.
- Bank-transfer reference.
- Nigerian address fields.
- Configurable tax rules.
- Common local units.
- Support for local onboarding and support channels.

Payments

The system should record:

- Cash.
- Bank transfer.
- POS terminal.
- USSD.
- Credit.

NIBSS Instant Payment supports real-time electronic transfers across multiple channels in Nigeria, including banking and digital payment channels. ^[8]

For subscription billing or online checkout, Paystack and Flutterwave can be integrated, but payment confirmation must be verified server-side rather than trusted solely from the browser. Paystack specifically documents verifying a transaction from the server using the transaction reference. ^[9]

12.27 Tax and e-invoicing readiness

The MVP should be tax-ready but should not make unsupported claims of full compliance.

Nigeria's electronic invoicing framework is evolving, and NRS resources describe a Merchant Buyer Solution intended to connect businesses and buyers and support fiscal compliance. ^[10]
^[11]

MVP requirements

- Configurable tax rates.
- Tax-inclusive prices.
- Tax-exclusive prices.
- Tax fields on receipts.
- Sequential invoice numbering.
- Customer tax information.
- Transaction export.
- Audit records.
- Integration-ready invoice data model.

Do not implement prematurely

- Unverified NRS claims.
- Unsupported compliance guarantees.
- Direct e-invoicing integrations before requirements are validated.
- Tax advice inside the product.

Future e-invoicing capability

Potential future requirements may include:

- Structured invoice submission.
- Invoice Reference Number.
- QR code.
- Taxpayer identification.
- Digital signatures.
- Credit notes.
- Debit notes.
- Submission status.
- Retry and rejection handling.

These requirements must be confirmed with official NRS documentation and qualified tax advisers before development.

13. Subscription and billing

Proposed plans

Free trial

- Limited duration.
- One branch.
- Basic products and sales.
- Limited number of transactions or products.

Starter

- One branch.
- Basic inventory and sales.
- Limited staff.
- Basic reports.

Growth

- More users.
- Customer and supplier balances.
- Stock counts.
- Advanced reports.
- More transaction volume.

Business

- Multiple branches.
- Advanced permissions.
- Transfers.
- Larger data limits.
- Priority support.

Pricing considerations

Local competitors advertise entry-level pricing from approximately ₦2,500 to ₦5,000 per month, while some offline products position annual plans near ₦120,000 per year. These figures should be treated as market signals rather than definitive pricing rules. ^[1] ^[2] ^[4]

The final pricing should be validated through customer interviews and paid pilots.

Billing requirements

- Display current plan.
- Start trial.
- Upgrade plan.
- Downgrade plan.
- Renew plan.
- Cancel plan.
- Show payment status.
- Handle payment failure.
- Provide grace period.
- Restrict features after grace period.
- Record billing events.
- Send subscription receipt.
- Show usage limits.

14. Technical requirements

Recommended architecture

Use a modular monolith for the MVP:

- Backend API.
- PostgreSQL database.
- Web dashboard.
- Mobile-friendly POS interface.
- Local offline database.
- Background job worker.
- Object storage.
- Monitoring.
- Automated backups.

This is more appropriate for an early-stage product than multiple microservices.

Shared platform modules

- Authentication.
- Organizations.
- Users.
- Roles.

- Permissions.
- Branches.
- Billing.
- Notifications.
- Audit logs.
- File storage.
- Support.

Inventory modules

- Products.
- Categories.
- Units.
- Suppliers.
- Customers.
- Stock balances.
- Stock movements.
- Purchases.
- Sales.
- Payments.
- Returns.
- Reports.
- Expenses.

Suggested technology options

Layer	Options
Frontend	React, Next.js, Vue
Mobile/POS	React Native, Flutter, PWA
Backend	Django, Laravel, NestJS, FastAPI
Database	PostgreSQL
Offline database	SQLite, IndexedDB, Dexie.js
Authentication	Supabase Auth, Firebase Auth, Auth.js, custom
Storage	S3-compatible storage
Payments	Paystack, Flutterwave
Hosting	Managed cloud platform

Layer	Options
Monitoring	Sentry or equivalent
Analytics	Privacy-conscious product analytics

Database entities

Platform

- organizations
- branches
- users
- memberships
- roles
- permissions
- subscriptions
- plans
- notifications
- audit_logs

Catalogue

- products
- product_variants
- categories
- brands
- units
- price_lists
- product_suppliers

Inventory

- stock_balances
- stock_movements
- stock_counts
- stock_count_lines
- stock_adjustments
- transfers
- transfer_lines

Sales

- sales
- sale_lines
- payments
- returns
- return_lines
- receipts

Purchasing

- purchase_orders
- purchase_lines
- goods_receipts
- supplier_payments

Relationships

- customers
- customer_payments
- customer_balances
- suppliers
- supplier_balances

Operations

- expenses
- imports
- exports
- sync_records
- integration_events

15. Data integrity rules

- Every tenant-owned record must include an organization ID.
- Every stock change must create a stock movement.
- Posted transactions cannot be silently edited.
- Corrections must use reversals or adjustments.
- Sale confirmation and stock deduction must be atomic.
- A received purchase must increase stock once.

- A draft purchase must not change stock.
- Duplicate sync requests must not create duplicate sales.
- Duplicate webhooks must not create duplicate payments.
- Archived products must remain available in history.
- Customer credit must update balances accurately.
- Returns must reference the original transaction where possible.
- All sensitive operations must generate audit records.

16. Offline and synchronization design

Local data

Cache only data required for daily operations:

- Active products.
- Prices.
- Stock snapshot.
- Customer lookup records.
- User permissions.
- Business configuration.
- Unsynchronized transactions.
- Recent local history.

Sync queue

Each transaction should contain:

- Local ID.
- Server ID after sync.
- Device ID.
- Organization ID.
- User ID.
- Timestamp.
- Payload.
- Status.
- Retry count.
- Error message.
- Idempotency key.

Sync strategy

- Use idempotency keys.
- Retry automatically.
- Use exponential backoff.
- Retain failed records.
- Make failures visible.
- Do not silently overwrite conflicts.
- Provide manual retry.
- Provide support diagnostics.

Offline limitations to disclose

- Cloud reports may be incomplete until synchronization.
- Two offline devices may sell the same remaining stock.
- Online payment verification may not work offline.
- Price data may become stale.
- Some administrative actions require internet access.

17. Security requirements

Authentication

- Secure password hashing.
- Session expiry.
- Token rotation.
- Password reset expiration.
- Rate limiting.
- Optional multi-factor authentication in a later release.

Authorization

- Server-side permissions.
- Organization isolation.
- Branch-level restrictions.
- Owner-controlled support access.
- Audit logging.

Data protection

- HTTPS/TLS.
- Secure secret storage.
- Encryption at rest where supported.
- Protected backups.
- No payment card data storage.
- Input validation.
- API rate limiting.
- Security monitoring.

Tenant isolation

Each query must be scoped to the correct organization. Use database row-level security or an equivalent robust mechanism.

Support access

Support access should be:

- Explicitly authorized.
- Time-limited.
- Logged.
- Restricted to necessary records.
- Visible to the owner where appropriate.

18. User experience requirements

Main navigation

1. Sell.
2. Products.
3. Stock.
4. Customers.
5. Reports.

Secondary menu:

- Purchases.
- Suppliers.
- Expenses.
- Staff.

- Settings.
- Billing.
- Support.

POS experience

The POS screen should prioritize:

- Search.
- Scan.
- Cart.
- Total.
- Payment.
- Complete sale.

A normal sale should not require the user to navigate through administrative settings.

Interface language

Use clear terms:

- Sell.
- Receive stock.
- Customer owes.
- Supplier balance.
- Stock correction.
- Low stock.
- Daily sales.
- Money received.

Avoid unnecessary ERP language.

Accessibility

- Large touch targets.
- Good contrast.
- Keyboard compatibility for desktop.
- Clear focus states.
- Readable font sizes.
- Error messages in plain language.
- No colour-only indicators.

19. Analytics and event tracking

Track:

- Account created.
- Organization created.
- Product added.
- Opening stock added.
- First sale.
- First offline sale.
- Sale synchronized.
- Sync failure.
- Receipt shared.
- Customer created.
- Credit sale.
- Customer payment.
- Purchase received.
- Stock adjustment.
- Stock count.
- Report viewed.
- CSV import.
- Subscription started.
- Payment completed.
- Subscription cancelled.

Avoid collecting unnecessary personal data.

20. Success metrics

Activation

- Percentage of businesses adding a product.
- Percentage entering opening stock.
- Percentage completing first sale.
- Time to first sale.

Engagement

- Weekly active businesses.
- Businesses recording sales weekly.
- Transactions per active business.
- Products per active business.
- Active users per business.
- Stock counts completed.
- Receipts shared.
- Credit records created.

Reliability

- Sync success rate.
- Duplicate transaction rate.
- Sync conflict rate.
- Crash-free sessions.
- Failed webhook rate.
- Stock discrepancy complaints.
- Number of lost or unrecoverable transactions.

Business

- Trial-to-paid conversion.
- Monthly recurring revenue.
- Churn.
- Average revenue per business.
- Customer acquisition cost.
- Referral rate.
- Support cost per business.

Customer value

- Reduction in stock discrepancies.
- Reduction in time spent on daily reporting.
- Increase in recorded sales.
- Increase in businesses using low-stock alerts.
- Owner confidence in business records.

21. Pilot plan

Pilot size

Start with 5–10 businesses in Enugu.

Pilot characteristics

Include:

- One single-location retailer.
- One business with multiple staff.
- One business with unstable internet.
- One business using credit sales.
- One business using barcode scanning.
- One business purchasing from several suppliers.

Pilot process

1. Conduct owner interview.
2. Observe current workflow.
3. Import products.
4. Configure opening stock.
5. Train owner.
6. Train staff.
7. Observe real transactions.
8. Monitor synchronization.
9. Review reports weekly.
10. Collect feedback.
11. Test payment willingness.
12. Identify repeated feature requests.

Pilot success criteria

- Users record real sales every week.
- Users trust stock numbers.
- Users continue after the first month.
- Users need less onboarding assistance over time.
- At least some users agree to pay.
- Sync and transaction errors remain low.

- Owners request additional staff or branches.

22. Development roadmap

Phase 0: Discovery

- Interview businesses.
- Select one launch vertical.
- Observe existing processes.
- Validate willingness to pay.
- Test prototypes.
- Review regulatory requirements.
- Confirm offline constraints.

Phase 1: Core product

- Authentication.
- Organization.
- Roles.
- Products.
- Categories.
- Opening stock.
- Sales.
- Basic stock movements.
- Basic dashboard.

Phase 2: Operational MVP

- Suppliers.
- Purchases.
- Customers.
- Credit sales.
- Payments.
- Receipts.
- Reports.
- CSV import/export.
- Low-stock alerts.
- Basic expenses.

Phase 3: Reliability

- Offline sales.
- Sync queue.
- Conflict handling.
- Returns.
- Stock counts.
- Audit logs.
- Backup and recovery.
- Better error reporting.

Phase 4: Commercial release

- Trial.
- Subscription plans.
- Payment integration.
- Usage limits.
- Billing page.
- Customer support workflow.
- Referral system.

Phase 5: Expansion

- Multi-branch transfers.
- Accounting.
- E-invoicing assessment.
- E-commerce.
- CRM.
- Staff management.
- Gym management.
- Bookings.

23. Risks and mitigation

Risk	Impact	Mitigation
MVP becomes too broad	Delayed launch	Enforce scope and prioritize daily workflows
Offline sync failure	Loss of customer trust	Use idempotency, retries, logs, and conflict handling
Users do not record every sale	Reports become unreliable	Make sales faster than manual entry

Risk	Impact	Mitigation
Difficult onboarding	Users abandon product	Provide assisted setup and CSV import
Low willingness to pay	Weak revenue	Test pricing during discovery and pilot
Incorrect profit values	Poor business decisions	Document costing and label estimates
Data breach	Severe reputation damage	Tenant isolation, encryption, permissions, backups
Regulatory changes	Compliance risk	Configurable tax model and professional review
Poor third-party modules	Security and upgrade problems	Review dependencies and use approved modules
Too many custom requests	High maintenance	Build common workflows, not one-off features
Duplicate payments	Financial disputes	Verify webhooks and use idempotency
Device loss	Data exposure	Cloud sync, session revocation, and access controls

24. Release readiness checklist

Product

- Core user flows tested.
- First-sale flow completed.
- Product import tested.
- Stock receiving tested.
- Credit sales tested.
- Returns tested.
- Reports reconciled.
- Offline flow tested.

Technical

- Tenant isolation tested.
- Permissions tested.
- Backups configured.
- Recovery tested.
- Sync retries tested.
- Duplicate prevention tested.
- Monitoring configured.
- Error logging configured.
- Payment webhooks verified.

Operational

- User onboarding guide ready.
- Help content ready.
- Support channel ready.
- Pricing confirmed.
- Terms and privacy policy ready.
- Pilot businesses recruited.
- Feedback process ready.

Compliance

- Business registration and operating structure reviewed.
- Data-protection obligations assessed.
- Tax claims reviewed.
- E-invoicing requirements investigated.
- Payment-provider terms reviewed.
- Customer data handling documented.

25. Product-manager recommendation

The first Orviohub product should not be marketed as a complete Nigerian Odoo replacement. It should be marketed as:

A simple, reliable inventory and sales system for Nigerian businesses.

The MVP should make this one process excellent:

Add product → receive stock → sell → record payment → issue receipt → reduce stock → view report → reorder.

If that workflow becomes part of a merchant's daily routine, Orviohub will have achieved product-market evidence. The future accounting, CRM, gym, booking, and business-platform modules should be built from that foundation only after customers consistently use and pay for the inventory product.



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